

YUBO ZHANG

Ph.D. Student in Electrical Engineering · Columbia University · Advisor: Prof. Xiaodong Wang

RESEARCH INTERESTS

Learning to optimize and in-context learning for wireless systems
Scalable and generalizable AI for large-scale MIMO
Integrated sensing, communication and backscatter systems
Communication security and optical wireless systems

EDUCATION

Columbia University	New York, NY Aug. 2024 – Present
Ph.D. in Electrical Engineering · GPA 4.0 / 4.3 · Advisor: Prof. Xiaodong Wang	
University of Science and Technology of China (USTC)	Hefei, China Sept. 2021 – Jul. 2024
M.Sc. in Information and Communication Engineering · GPA 3.6 / 4.3 · Advisor: Prof. Chen Gong	
University of Science and Technology of China (USTC)	Hefei, China Sept. 2017 – Jul. 2021
B.Eng. in Electronic Information Engineering · GPA 3.7 / 4.3, ranked top 10% of 313 students	

RESEARCH & PROJECT EXPERIENCE

Learning Transition Policies for High-Order MIMO Reception	Apr. 2026 – Present
<i>Research Assistant, Columbia University · with Prof. Xiaodong Wang and Yiyao Liu</i>	
- Reframed high-order MIMO detection as reinforcement learning over transitions between complete transmit vectors, and designed a channel-coupled Transformer with blockwise autoregressive sampling to model dense inter-stream dependence while searching exponentially large QAM spaces. - Developed a staged hard-to-soft receiver pipeline: residual-to-BER curriculum training, checkpoint transfer to soft-input/soft-output iterative detection and decoding (IDD), decoder-prior-conditioned proposal sampling, likelihood-weighted posterior/extrinsic LLR construction, and decoder-in-the-loop fine-tuning. - Extended the receiver to pilot-only operation without explicit channel estimation by building an in-context learner over protocol-provided pilot pairs; implemented the end-to-end PyTorch training and evaluation stack, including proposal-policy transfer, likelihood-energy calibration, and conservative context evolution.	
Physics-Grounded Cross-Modal Pretraining for Camera-Channel ISAC	Jul. 2026 – Present
<i>Research Mentor, Columbia University · with Yiyao Liu</i>	
- Architected a lightweight ViT-based self-supervised framework for synchronized camera images and noisy wideband CSI, transforming channels into 3D delay-angle beamspace tokens and learning reusable representations with limited task labels. - Combined soft contrastive alignment, physics-guided asymmetric cross-modal masking, and entropic optimal-transport correspondence to capture fine-grained path-patch relationships without relying on a large vision-language backbone. - Designed the four-stage pretraining curriculum, ablation matrix, and reproducible downstream pipeline for localization, LoS/NLoS classification, beam selection, blockage detection, and rate prediction, with frozen-encoder few-shot transfer and single-GPU efficiency as core design constraints.	

Learning-to-Anneal for Graph Combinatorial Optimization

2026 – Present

Research Assistant, Columbia University

- Formulated a population-based reinforcement-learning solver that amortizes NP-hard graph optimization across instance distributions and recursively transitions between complete solutions, rather than relying on one-shot construction or hand-designed local moves.
- Developed trajectory policy-gradient and actor-critic training, blockwise autoregressive and random-masking samplers, and an extended Transformer that jointly updates candidate solutions and graph embeddings while retaining global node dependencies.
- Designed objective-smoothing and graph-scale curricula to stabilize large-graph training, and implemented Max-Cut evaluations against learning-based baselines and time-limited Gurobi with emphasis on low-latency inference and cross-size generalization.

Self-Evolving In-Context Learning for Direct Pilot-to-Beamformer Design

Nov. 2025 – Apr. 2026

Research Assistant, Columbia University · with Prof. Xiaodong Wang

- Built an in-context learning framework that maps uplink pilot observations directly to downlink beamformers in MU-MISO systems, removing the explicit channel-estimation stage from the design pipeline.
- Designed a self-evolving training loop in which the model progressively refines its own in-context examples, so that the learned mapping improves without additional labeled supervision.
- Implemented the full Transformer-based architecture and training pipeline in PyTorch; the model sustains sum-rate performance across varying user counts and pilot lengths with no retraining.

SALLO-M: Transformer Model for Scalable and Generalizable MU-MISO Beamforming

Apr. 2025 – Oct. 2025

Research Assistant, Columbia University · with Dr. Xiao-Yang Liu

- Designed and trained a multi-layer Transformer model from scratch for large-scale MU-MISO beamforming optimization, integrating curriculum learning and semi-amortized learning strategies.
- Implemented the full architecture and training pipeline in PyTorch, achieving 5% higher sum-rate than the WMMSE algorithm with only 3% of its inference time, while retaining strong generalization to dynamic communication scenarios.
- Employed random masked attention to improve scalability and robustness across antenna and user configurations, and designed trainable sparse attention for higher training efficiency.

Encoder-Decoder Network for FDD Large-Scale Sparse Channel Precoding

Sept. 2024 – Apr. 2025

Research Assistant, Columbia University · published in IEEE OJCOMS

- Designed and trained an encoder-decoder network for joint channel recovery and beamforming design under large-scale sparse channels, and extended the framework to hybrid beamforming in both far-field and near-field regimes.
- Completed the problem formulation, network architecture, training algorithm and PyTorch implementation, addressing local-optima entrapment and limited feedback overhead.
- The resulting model outperformed MMSE baselines by 13%–37% across channel models and SNR settings.

Deep Reinforcement Learning for Keyless Positive-Rate Covert Communication

Feb. 2024 – Aug. 2024

Research Assistant, Columbia University

- Developed an MDP model and trained a deep Q-network tailored to causal power- and rate-allocation problems, and proposed multi-step non-convex optimization schemes for the non-causal case.
- Completed the optimization and DQN framework design and its PyTorch implementation; the DQN approach improved covert-rate performance by 10%–26% over average-allocation schemes in block-fading channels.

Direct and Ambient Backscatter Communications with a Dual-Function Radar Transmitter

Apr. 2025 – Feb. 2026

Research Assistant, Columbia University · with Prof. Luca Venturino

- Modeled direct and ambient backscatter links illuminated by a dual-function radar transmitter, and derived the corresponding detection and communication performance metrics for both regimes.
- Optimized transmit resources jointly for sensing and backscatter communication, characterizing the achievable trade-off between radar detection quality and backscatter throughput.

1 Mbps Multi-User Ad-Hoc Ultraviolet Optical Communication System on FPGA

Apr. 2023 – Sept. 2023

Hardware & Software Engineer, USTC

- Achieved network synchronization using beacon sequences, designed the frame structure, and developed a multi-transceiver TDMA communication network.
- Integrated the physical- and MAC-layer modules, conducted system-level testing, and completed the FPGA development and debugging of the synchronization and TDMA modules.

Graph-Based Conflict-Free MAC Protocol for a Two-Layer Communication Network

Dec. 2022 – Jun. 2023

Research Assistant, USTC

- Designed a resource-allocation scheme for a two-layer network, analyzed link interference using a conflict graph and Monte-Carlo simulation, and optimized the routing algorithm.
- Contributed the interference-model analysis, routing-algorithm design and data-fitting work for the resulting protocol.

Multi-User Uplink Communication with a Spatial-Multiplexing Receiver

Sept. 2022 – Mar. 2023

Research Assistant, USTC

- Designed pilot sequences from an analytical MSE objective, analyzed the Gaussian approximation, and adopted successive-elimination detection (SED) to handle unknown interference.
- Implemented the estimation and detection schemes in MATLAB: the pilot design approached the optimal estimation MSE, and SED achieved near-ML BER at 5%–10% of the execution time.

TDOA-Based Ultraviolet Positioning System on FPGA

May 2021 – Sept. 2021

Hardware Engineer, USTC

- Developed a TDOA-based ultraviolet positioning system and conducted large-scale outdoor field tests.
- Responsible for the system-level code implementation, data acquisition and performance validation.

10 kbps Multi-Service Real-Time Ultraviolet Communication System on FPGA

May 2021 – Sept. 2021

Software & Test Engineer, USTC

- Developed an FPGA-based ultraviolet communication system with a custom synchronization scheme and frame structure, supporting simultaneous real-time video, audio and text transmission.
- Completed the coding for audio and text transmission and coordinated system integration and testing.

PUBLICATIONS

1. **Yubo Zhang** and Xiaodong Wang. "Self-Evolving In-Context Learning for Direct Pilot-to-Beamformer Design in MU-MISO Systems." *ArXiv preprint*, 2026.
2. **Yubo Zhang**, Luca Venturino and Xiaodong Wang. "Direct and Ambient Backscatter Communications with a Dual-Function Radar Transmitter." *ArXiv preprint*, 2026.
3. **Yubo Zhang**, Jeremy Johnston and Xiaodong Wang. "Semi-Amortized Encoder-Decoder Network for Beamforming over Sparse Large-Scale MIMO Channels." *IEEE Open Journal of the Communications Society*, 2026.
4. **Yubo Zhang**, Xiao-Yang Liu and Xiaodong Wang. "A Semi-Amortized Lifted Learning-to-Optimize Masked (SALLO-M) Transformer Model for Scalable and Generalizable Beamforming." *ArXiv preprint*, 2025.
5. **Yubo Zhang**, Hassan ZivariFard and Xiaodong Wang. "Resource Allocation for Positive-Rate Covert Communications Using Optimization and Deep Reinforcement Learning." *ArXiv preprint*, 2025 (*IEEE Trans. Wireless Communications*, in revision).
6. Yuchen Pan, **Yubo Zhang**, Shihui Yu, Junfan Chen, Chen Gong and Zhengyuan Xu. "Beacon-Based Time Synchronization Protocol Design for Layered Ultraviolet Communication Network." *IEEE Trans. Communications*, 2025.
7. Yifan Ding, Chen Gong, Weijie Liu, **Yubo Zhang**, Haidong Sun and Zhengyuan Xu. "Multi-Input UV Communication with Adaptive Gain Adjustment." *Journal of Lightwave Technology*, 43(3): 1170–1178, 2025.

8. **Yubo Zhang**, Yuchen Pan, Chen Gong, Beiyuan Liu and Zhengyuan Xu. "Channel Estimation and Signal Detection for NLOS Ultraviolet Scattering Communication with Space Division Multiple Access." *IEEE Trans. Communications*, 72(10): 6427–6441, 2024.
9. Shihui Yu, Chubing Lv, Yueke Yang, Yuchen Pan, Lei Sun, **Yubo Zhang**, Juliang Cao, Ruihang Yu, Chen Gong, Wenqi Wu and Zhengyuan Xu. "Ultraviolet Positioning via TDOA: Error Analysis and System Prototype." *Optics Express*, 32(19): 33144, 2024.
10. Yuchen Pan, **Yubo Zhang**, Fei Long, Ping Li, Haotian Shi, Jiazhao Shi, Hanlin Xiao, Chen Gong and Zhengyuan Xu. "Beacon-Enabled TDMA Ultraviolet Communication Network Design and Realization." *CSNDSP*, 2024.
11. Yuchen Pan, Guanchu Wang, **Yubo Zhang**, Jingyin Tang, Chen Gong and Zhengyuan Xu. "Graph-Based Conflict-Free MAC Protocol and Conflict Analysis for a Two-Layer Ultraviolet Communication Network." *JOCN*, 15(6): 381–392, 2023.
12. Yifan Ding, **Yubo Zhang**, Huabin Yu, Chen Gong, Haiding Sun and Zhengyuan Xu. "266 nm Ultraviolet Communication under Unknown Interference Using UVC Micro-LED." *Optics Express*, 31(10): 16406–16422, 2023.
13. **Yubo Zhang**, Chen Gong and Zhengyuan Xu. "Bi-Directional Ultraviolet Communication with Self-Interference." *Optics Express*, 30(21): 38534–38549, 2022.

SKILLS, AWARDS & COURSEWORK

Programming

Python (6 yrs) · PyTorch (4 yrs) · MATLAB (6 yrs) · C (4 yrs) · C++ (2 yrs) · Verilog (4 yrs)

Awards

G.D.C. Corporation Scholarship (top 2%) · Outstanding Graduate of USTC (top 10%) · Excellent Student of USTC, Silver Award (top 10%, two consecutive years)

Coursework

Optimization (100/100) · Stochastic Processes (97/100) · Optical Networks (97/100)